

Topic 5

Preference heterogeneity

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Introduction

Motivation

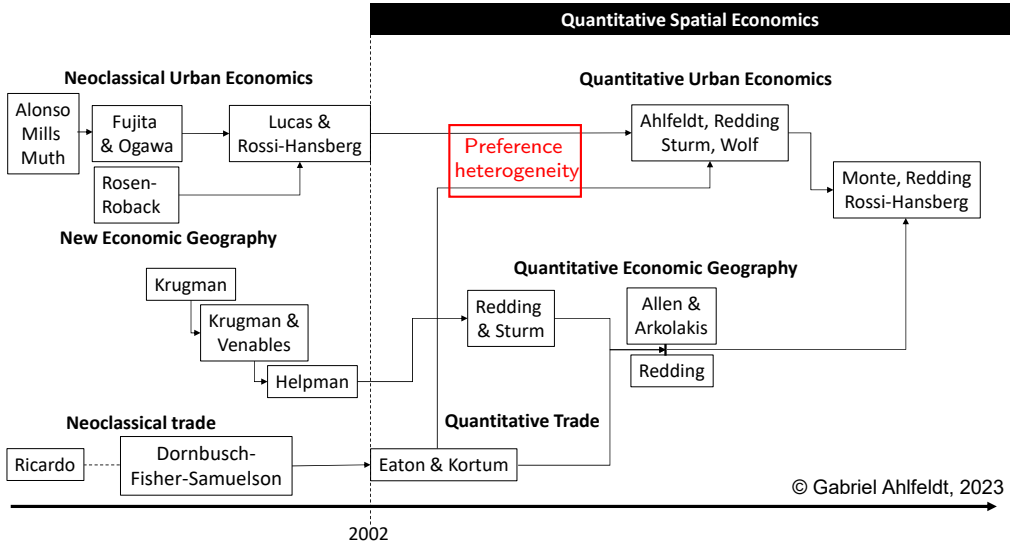
- ▶ So far, individuals have been **homogenous**
 - ▶ All workers value place-specific amenities the same
 - ▶ Infinite supply of workers to any location for a given combination of
 - ▶ Wages
 - ▶ Prices (including rents)
 - ▶ Place attributes
- ▶ In QSMs, we (usually) add **idiosyncratic tastes for location**
 - ▶ Drawn from extreme value distributions

What do idiosyncratic tastes capture and why do they matter?

Roadmap

- ▶ **Review extreme-value distributions**
 - ▶ Fréchet and Gumble
- ▶ **Derive location choice probabilities**
 - ▶ Binary case
 - ▶ Multiple locations
- ▶ **Role of preference heterogeneity in the spatial general equilibrium**
 - ▶ Dispersion of idiosyncratic tastes affects how
 - ▶ differences in fundamentals lead to
 - ▶ differences in prices and quantities on labour and land markets

History of thought



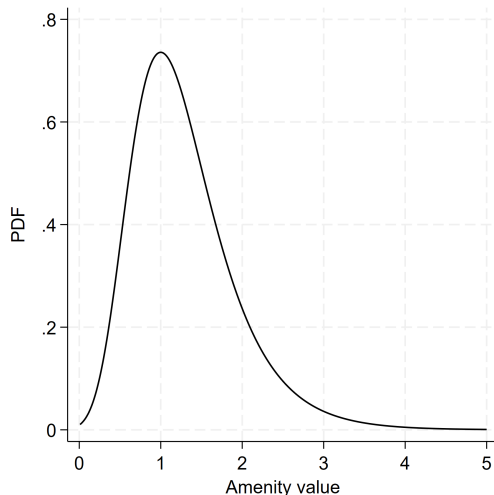
Extreme value distributions

Extreme value type I (Gumbel)

$$g(a) = \frac{1}{\beta} \exp \left(- \left[\frac{a-A}{\beta} + \exp \left(- \frac{a-A}{\beta} \right) \right] \right)$$

- ▶ We draw the value of the idiosyncratic amenity a from a distribution $g(a)$
 - ▶ First moment:
 $E(a) = A + \beta\Gamma = A + \left(\frac{1}{\varepsilon}\right)\Gamma$
 - ▶ Second moment:
 $\sigma_a^2 = \frac{\pi^2}{6}\beta^2 = \frac{\pi^2}{6}\left(\frac{1}{\varepsilon}\right)^2$
 - ▶ where $\varepsilon \equiv 1/\beta$ and Γ is the Euler–Mascheroni constant

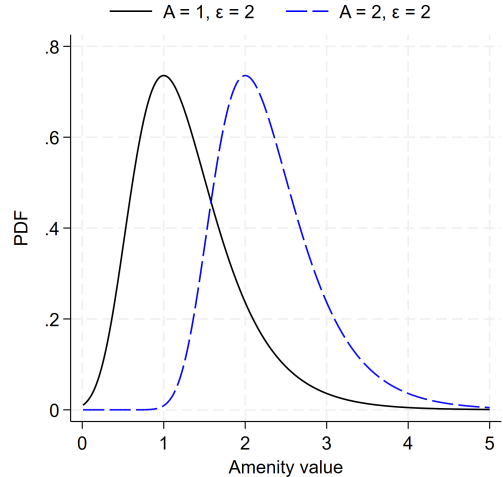
What happens if we increase A or ε ?
What does this mean intuitively?



Extreme value type I (Gumbel)

$$g(a) = \frac{1}{\beta} \exp \left(- \left[\frac{a-A}{\beta} + \exp \left(- \frac{a-A}{\beta} \right) \right] \right)$$

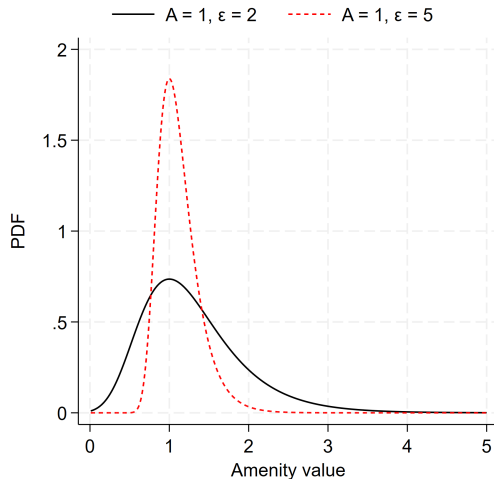
- ▶ A governs the first moment
 - ▶ E.g. the average idiosyncratic amenity at a location
- ▶ **Increasing A implies a higher amenity, on average**



Extreme value type I (Gumbel)

$$g(a) = \frac{1}{\beta} \exp \left(- \left[\frac{a-A}{\beta} + \exp \left(- \frac{a-A}{\beta} \right) \right] \right)$$

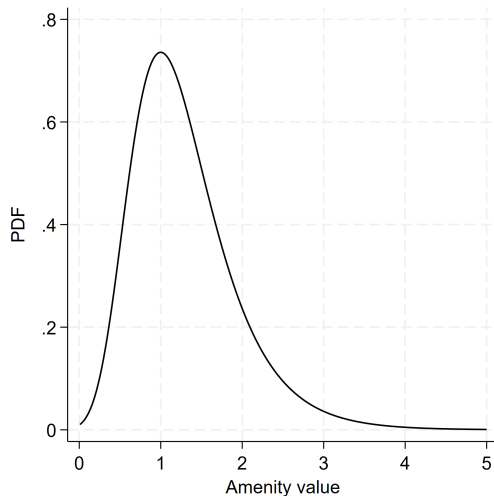
- ▶ ε affects first and second moments
 - ▶ the average idiosyncratic amenity
 - ▶ the dispersion
- ▶ **Increasing ε makes workers more similar in tastes**
 - ▶ less dispersion in tastes



Extreme value type II (Fréchet)

$$f(a) = \frac{\varepsilon}{A} \left(\frac{a}{A}\right)^{-1-\varepsilon} \exp\left(-\left(\frac{a}{A}\right)^{-\varepsilon}\right)$$

- ▶ We draw the value of the idiosyncratic amenity a from a distribution $f(a)$
 - ▶ First moment: $E(a) = A\Gamma\left(1 - \frac{1}{\varepsilon}\right)$
 - ▶ Second moment: $\sigma_a = A^2\Gamma\left(1 - \frac{2}{\varepsilon}\right) - \left(\Gamma\left(1 - \frac{1}{\varepsilon}\right)\right)^2$
 - ▶ where Γ is the Euler–Mascheroni constant
- ▶ Comparative statics similar to Gumbel
 - ▶ Larger $A \Rightarrow$ greater $E(a)$
 - ▶ Larger $\varepsilon \Rightarrow$ smaller σ_a
 - ▶ Though A and ε affect $E(a)$ and σ_a



Lessons for inversion

► Recall

- QSM has parameters and fundamentals
- For given parameters we invert first moment of fundamental (e.g. average amenity)

► Notice

- First moment depends on structural parameter ε
- Under Gumbel and Fréchet

► Remember

- Fundamental amenity recovered is always specific to the chosen value of ε
- Therefore, cannot change ε without changing A

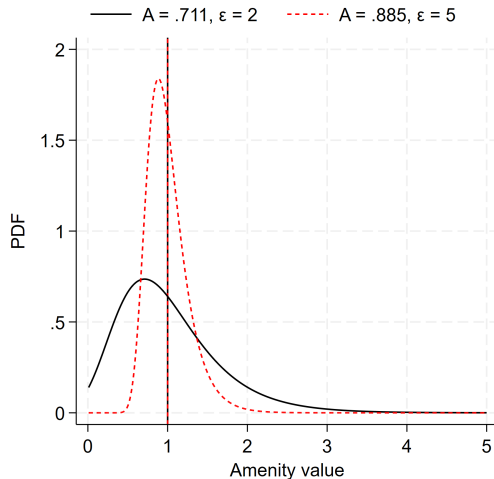
**Can do counterfactuals under different values of ε .
But predicting what happens when ε changes is not straightforward**

Inverting A (Gumbel)

$$A = E(a) - \frac{1}{\varepsilon} \Gamma$$

- ▶ We re-arrange first moment to find A
 - ▶ Can rationalize any given $E(a)$
- ▶ Say, we want $E(a) = 1$
 - ▶ We set $A = 0.711$ if $\varepsilon = 2$
 - ▶ We set $A = 0.885$ if $\varepsilon = 5$
- ▶ Principle is the same with Fréchet
 - ▶ Just with a different formula

$$A = \frac{E(a)}{\Gamma(1 - \frac{1}{\varepsilon})}$$



Location choice probabilities

Discrete choice

- ▶ We want to move **beyond perfect mobility**
 - ▶ infinite supply of homogeneous workers
- ▶ Preference heterogeneity generates **well-behaved location choice probabilities**
 - ▶ Some people will **always choose a location**, even if real wage and amenity are low
 - ▶ But it could be a very small fraction of the population (with extreme preferences)
 - ▶ A **better location attracts a greater share** of the worker endowment, $\bar{L}$
 - ▶ Better in terms of higher wages, lower rents, lower prices, higher amenities
- ▶ Application of McFadden's (1974) seminal work
 - ▶ Nobel Prize winner in 2000

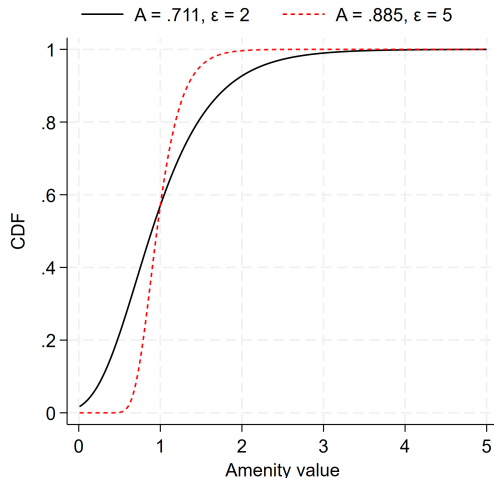
Let's develop the intuition in a simple binary choice setting: City vs. RoW

Cumulative density function (Gumbel)

$$G(a) = \exp(-\exp(-(a - A)\varepsilon))$$

- ▶ CDF tells us the probability that a is equal or smaller than a certain value
 - ▶ Let a be idiosyncratic urban utility
 - ▶ About 60% chance that $a \leq 1$
- ▶ Outside option offering utility $\bar{U} = 1$
 - ▶ Probability of **not choosing city**:
 $Pr(a \leq \bar{U}) = G(\bar{U}) \approx 60\%$
 - ▶ Probability of **choosing city**:
 $Pr(a > \bar{U}) = 1 - G(\bar{U}) \approx 40\%$

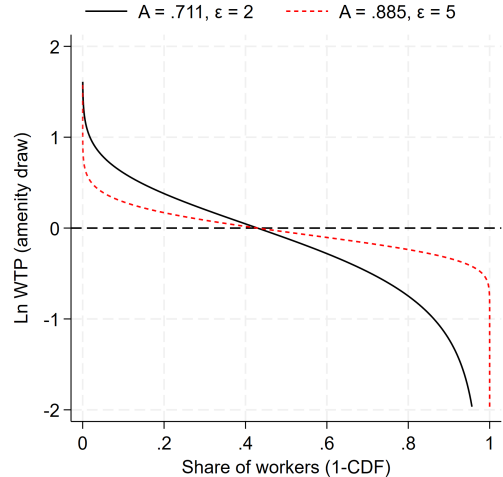
CDF gives us choice probabilities!



Demand for residence

- CDF implies **downward-sloping demand for residence**
 - Number of workers in the city is $L = \mu \bar{L}$, where $\mu = 1 - G(\bar{U}) \approx 40\%$

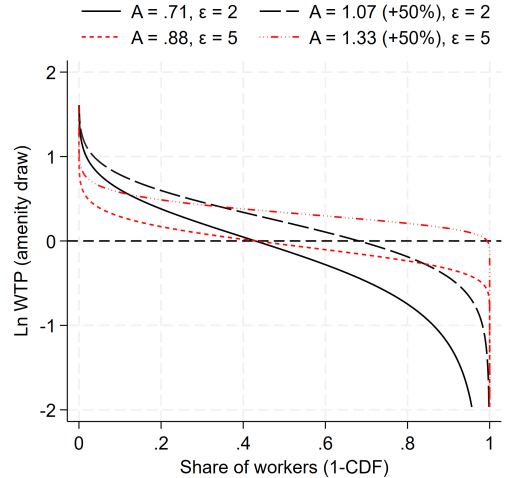
Valuation of the marginal resident falls the more workers enter the city!



Role of ε

- ▶ If we **increase** A , the residence demand curve shifts outwards
 - ▶ Average utility increases
 - ▶ **City attracts workers who enjoy urban life less relative to others**
- ▶ 50% increase in A leads to an increase in urbanization rate from $\approx 40\%$
 - ▶ to $\approx 70\%$ if $\varepsilon = 2$
 - ▶ close to 100% if $\varepsilon = 5$

ε governs population response in counterfactuals!



Multi-location setting I

- ▶ The logic extends to the setting with many locations (maths more involved)
 - ▶ **You will need to go through the intermediate steps at your own pace** (tutorial)
- ▶ Let's now assume workers indexed by o
 - ▶ can choose among many locations i
 - ▶ receive idiosyncratic utility according to Fréchet CDF $F(a_o) = \exp(-a_o^{-\epsilon})$
 - ▶ obtain indirect utility $U = A_i a_{io} \frac{w_i}{p_i^{1-\alpha}}$
- ▶ We need: The share of workers getting a utility of up to u at i is given by CDF
 - ▶ $G_i(u) = \Pr(U \leq u) = \Pr\left(A_i a_{io} \frac{w_i}{p_i^{1-\alpha}} \leq u\right)$
- ▶ Rearrange the $\Pr(\cdot)$ term to be given in terms of shock a so we can use $F(a)$
 - ▶ $\Pr\left(a_{io} \leq \frac{p_i^{1-\alpha}}{w_i A_i} u\right) = F\left(\frac{p_i^{1-\alpha}}{w_i A_i} u\right) \Rightarrow G_i(u) = \exp\left(-\left(\frac{p_i^{1-\alpha}}{A_i w_i}\right)^{-\epsilon} u^{-\epsilon}\right)$

Multi-location setting II

- ▶ Want to know the **probability that workers choose i** , $\pi_i = Pr(u_i \geq \max\{u_r\} \forall i)$
 - ▶ Intuitively, this is the **probability that workers achieve a utility u at i** , $g_i(u)$, multiplied by the **probability that the maximum utility in any other location is less than or equal to u** , $\prod_{r \neq i} G_r(u)$, evaluated over all u :
 - ▶ $\pi_i = \int_0^\infty g_i(u) \left(\prod_{r \neq i} G_r(u) \right) du$
- ▶ Several steps later (using $G(u)$ from which we also derive $g(u)$...)
 - ▶ **Location choice probability** $\mu_i = \frac{L_i}{L} = \frac{\left(\frac{A_i w_i}{p_i^{1-\alpha}} \right)^\epsilon}{\left(\sum_s \left(\frac{A_s w_s}{p_s^{1-\alpha}} \right)^\epsilon \right)}$

Eureka! ϵ is the labour supply elasticity to the city! (Moretti, 2010)

Role in general equilibrium

Implications for quality of life

- ▶ Recall **quality of life in Rosen-Roback framework** (Glaeser & Gottlieb, 2009)
 - ▶ $A_i = c \frac{p_i^{1-\alpha}}{w_i}$ (in GG2009 notations $\log(B_c) = \Omega_3 + \sigma \log(P_c) - \log(W_c)$)
- ▶ Solving the **location choice probability** equation for A_i , we get
 - ▶ $A_i = c \underbrace{\frac{p_i^{1-\alpha}}{w_i}}_{\text{RR-QoL}} L^{\frac{1}{\varepsilon}}$ Notice that if $\varepsilon \rightarrow \infty \Rightarrow L^{\frac{1}{\varepsilon}} \rightarrow 1$
- ▶ **QSM nests Rosen-Roback as special case** with $\varepsilon \rightarrow \infty$
 - ▶ By implication: If $\varepsilon < \infty$ and QSM is right, Rosen-Roback must give wrong QoL

Does RR over- or underestimate the urban QoL premium?

Ahlfeldt, Bald, Roth, Seidel toolkit

- ▶ **Toy version** of the full quantitative spatial model
 - ▶ Abstracts from local ties, trade costs and non-tradable sector in the DP
 - ▶ Careful, *local ties are really important!* [full paper](#)
 - ▶ But offers **analytical solutions!**
- ▶ Get toolkit
 - ▶ GitHub repository: [▶ Open ABRS-toolkit](#)
 - ▶ Contains the description of the [toy model and intuition](#), **worth reading**
 - ▶ Install Stata ado file: `ssc install ABRS`
 - ▶ Creates the four-quadrant diagram
 - ▶ For the syntax, type: `help ABRS`
 - ▶ Javascript version: [▶ Open ABRS-toolkit \(web-version\)](#)
 - ▶ Only bar chart showing effects on equilibrium outcomes

Preferences

- ▶ Worker living in location i derives utility from the consumption of goods ($Q_{i\omega}$) and floor space ($h_{i\omega}$) according to

$$U_{i\omega} = \left(\frac{Q_{i\omega}}{\alpha} \right)^\alpha \left(\frac{h_{i\omega}}{1-\alpha} \right)^{1-\alpha} \exp[a_{i\omega}], \quad (1)$$

- ▶ where $Q_{i\omega}$ represents a final good (Armington, 1969)
 - ▶ locally assembled from tradable intermediate goods $q_{ji\omega}$
 - ▶ according to the CES-aggregator
 - ▶ at zero cost under perfect competition
 - ▶ shipped from origin j to destination i ,

$$Q_{i\omega} = \left[\sum_{j \in J} (q_{ji\omega})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (2)$$

Mobility frictions

- ▶ Idiosyncratic preference for location $\exp[a_{i\omega}^\theta]$ (Gumbel)

$$F_i^\theta(a) = \exp\left(-(A_i)^{\gamma^\theta} \exp\{-[\gamma a + \Gamma]\}\right) \quad \text{with } \gamma > 0, \quad (3)$$

- ▶ A_i (average preference) serves as an exogenous measure of local quality of life
 - ▶ Notice that **Gumbel CDF has been engineered** to give $A_i = \mathbb{E}(a|i)$
 - ▶ Using $-\Gamma$ and $(A_i)^\gamma$ ensures $\mathbb{E}[a_i] = \ln A_i$, independent of γ (credit to Jens Wrona)
- ▶ γ governs the dispersion of individual amenity shocks
 - ▶ Introduces imperfect spatial arbitrage $\Rightarrow$
 - ▶ inverse measure of mobility frictions

Technology: Housing

- ▶ Supplied under **perfect competition** combining a share of the globally available capital stock, K_i (available at unit prices), with **fixed land supply**, $\bar{T}_i$:

$$H_i^S = \eta_i \left(\frac{\bar{T}_i}{\delta} \right)^\delta \left(\frac{K_i}{1-\delta} \right)^{1-\delta}. \quad (4)$$

- ▶ Total factor productivity η_i
 - ▶ captures the role of regulatory (e.g. height regulations) and physical (e.g. a rugged surface) **constraints** (Saiz 2010)
- ▶ δ governs the housing supply elasticity $\frac{\partial \ln H_i^S}{\partial \ln p_i^H} = \frac{1-\delta}{\delta}$ (inverse relationship)
 - ▶ Use first-order conditions in profit function
- ▶ Generates **upward-sloping housing supply** (similar to Glaeser & Gottlieb, 2009)

Technology: Non-housing goods

- ▶ **Each location produces a unique variety** of a tradable intermediate good
 - ▶ using labour L_i as the only production input
 - ▶ under perfect competition according to
- ▶ $q_i = \varphi_i L_i$
 - ▶ Endogenous labour productivity $\varphi_i = \bar{\varphi}_i L_i^\zeta$
 - ▶ increases in local employment according to the **agglomeration elasticity** ζ .
- ▶ Expenditure share $\chi_{ji} = \frac{(w_j/\varphi_j)^{1-\sigma}}{\sum_k (w_k/\varphi_k)^{1-\sigma}}$ ((imports from j to i)
 - ▶ Trade in intermediate goods is free
 - ▶ Perfect competition equates prices to marginal costs (price $p_{ji} = w_j/\varphi_j$)
- ▶ Generates **downward-sloping labour demand**

Location choice

- Probability λ_i^θ that a worker lives in location i :

$$\lambda_i = \frac{(A_i w_i / \mathcal{P}_i)^\gamma}{\sum_{j \in J} (A_j w_j^\theta / \mathcal{P}_j)^\gamma}, \quad (5)$$

- where $\mathcal{P}_i \equiv (P_i)^\alpha (p_i^H)^{1-\alpha}$ is the aggregate consumer price index.
 - Mobility of workers equalizes expected utility in equilibrium
- **Upward-sloping labour supply $\Rightarrow$ downward-sloping housing demand**

This is what we have derived before...

General equilibrium I

- ▶ **Goods market clearing (GMC):** $w_i L_i = \sum_{j \in J} \frac{(w_i / \varphi_i)^{1-\sigma}}{\sum_k (w_k / \varphi_k)^{1-\sigma}} (\alpha w_j L_j + r_j \bar{T}_j)$
 - ▶ Wage bill at i must equate to the revenues from all locations j
 - ▶ (Firm) *income equals expenditure*
- ▶ **Labor market clearing (LMC):** $L_i = \lambda_i \bar{L}$
 - ▶ Labour input L_i must equate labour supply
 - ▶ product of the share of workers living in i , λ_i , and the national labour endowment $\bar{L}$.
- ▶ **Floor-space market clearing (FMC):** $p_i^H = \left(\frac{\tilde{\alpha} \delta w_i L_i}{\eta_i^{\frac{1}{\delta}} \bar{T}_i} \right)^\delta$,
 - ▶ where $\tilde{\alpha} \equiv (1 - \alpha) + \alpha(1 - \beta)(1 - \delta)$ is a constant.
 - ▶ Simply set $H_i^S = H_i^D$
 - ▶ H^D is Marshallian housing demand
 - ▶ $H_i^S = \eta_i^{\frac{1}{\delta}} \bar{T}_i (p_i^H)^{\frac{1-\delta}{\delta}} / \delta$ from first-order conditions in profit function

General equilibrium II

- ▶ For the toolkit, we consider **relative differences in a two-region case**

- ▶ $\hat{x} = \frac{x_i}{x_j}$

- ▶ System of equations simplifies to

- ▶ GMC: $\hat{L} = \hat{\varphi}^{\frac{\sigma-1}{\Lambda}} \hat{w}^{-\frac{\sigma}{\Lambda}}$

- ▶ LMC: $\hat{L} = \left(\hat{A} \hat{w} (\hat{p}^H)^{\alpha-1} \right)^{\gamma}$

- ▶ FMC: $\hat{p}^H = \hat{k} \left(\hat{w} \hat{L} \right)^{\delta}$

- ▶ where $\Lambda \equiv 1 - \zeta(\sigma - 1)$ and $k_i \equiv \frac{1}{\eta_i} \left(\frac{\tilde{\alpha}\delta}{T_i} \right)^{\delta}$ collect various primitives

3 log-linear equations with 3 unknowns $\{\hat{L}, \hat{w}, \hat{p}^H\} \Rightarrow$ **closed-form solution** ✓

Mapping

- We have **direct mapping from primitives to endogenous objects**

$$\hat{L} = \hat{k}^{-\frac{(1-\alpha)\sigma}{\Delta}} \hat{\varphi}^{\frac{(\sigma-1)(1-(1-\alpha)\delta)}{\Delta}} \hat{A}^{\frac{\sigma}{\Delta}}$$

$$\hat{w} = \hat{k}^{\frac{(1-\alpha)\Lambda}{\Delta}} \hat{\varphi}^{\frac{(\sigma-1)(1/\gamma+(1-\alpha)\delta)}{\Delta}} \hat{A}^{-\frac{\Lambda}{\Delta}}$$

$$\hat{p}^H = \hat{k}^{\frac{\Lambda+\sigma/\gamma}{\Delta}} \hat{\varphi}^{\frac{(\sigma-1)(1+1/\gamma)\delta}{\Delta}} \hat{A}^{\frac{\delta(\sigma-\Lambda)}{\Delta}}$$

- where $\Delta \equiv \Lambda[1 - (1 - \alpha)\delta] + \sigma[1/\gamma + (1 - \alpha)\delta]$
- This mapping is all we need to do counterfactuals in spatial general equilibrium
 - change fundamentals $\{A, \varphi, \eta\}$ [► Open ABRS-toolkit \(web-version\)](#)

GE in four quadrants

- For **didactic purposes** it is useful to combine GMC, LMC, FMC to give

$$\hat{L} = \hat{k}^{-\frac{1-\alpha}{\frac{1}{\gamma}+(1-\alpha)\delta}} \cdot \hat{A}_1^{\frac{1}{\frac{1}{\gamma}+\delta(1-\alpha)}} \cdot \hat{w}^{\frac{1-\delta(1-\alpha)}{\frac{1}{\gamma}+\delta(1-\alpha)}}$$

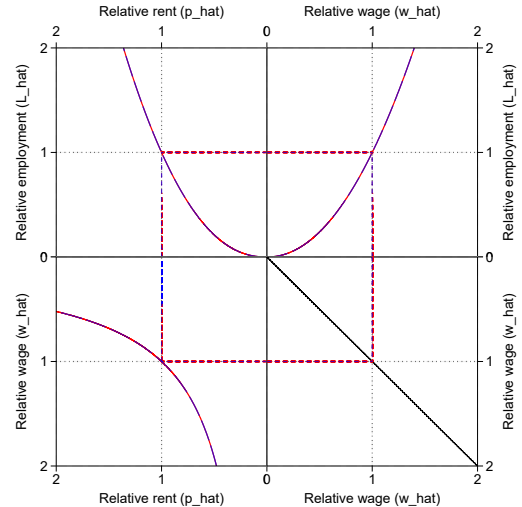
$$\hat{p}^H = \hat{k}^{\frac{1}{1-(1-\alpha)\delta}} \cdot \hat{A}_1^{-\frac{\delta}{1-\delta(1-\alpha)}} \cdot \hat{L}^{\frac{\delta(\frac{1}{\gamma}+1)}{1-(1-\alpha)\delta}}$$

$$\hat{w} = \left(\frac{\hat{p}^H}{\hat{k}} \right)^{\frac{\zeta(\sigma-1)-1}{\delta(\sigma-1)(\zeta+1)}} \cdot \hat{\varphi}^{\frac{1}{\zeta+1}}$$

- Notice that we map $\hat{w}$ into $\hat{L}$, $\hat{L}$ into $\hat{p}^H$, and then $\hat{p}^H$ back into $\hat{w}$
- A $\hat{w}$ that gets us back to the same $\hat{w}$ satisfies all equilibrium conditions

ABRS " $\hat{A}_1=0$ "

- ▶ Quadrant 1
 - ▶ Higher worker wage $\Rightarrow$ greater employment due to **upward-sloping labour supply**
- ▶ Quadrant 2
 - ▶ Greater employment $\Rightarrow$ higher rent due to greater housing demand & **imperfectly elastic supply**
- ▶ Quadrant 3
 - ▶ Higher rent implies larger employment $\Rightarrow$ Firm wage must be lower given **downward-sloping labour demand**
- ▶ Quadrant 4
 - ▶ Projects firm wage onto worker wage



Questions I

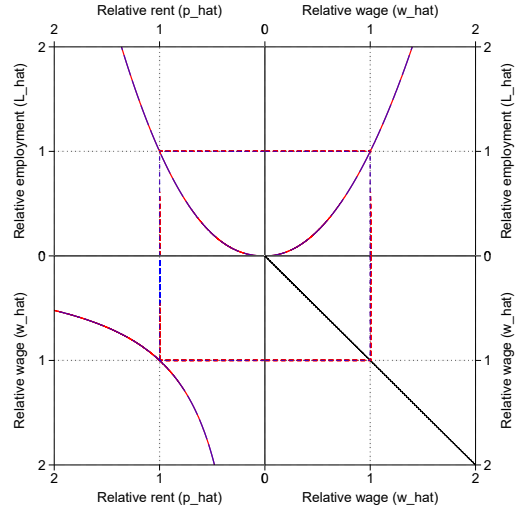
Q1: What happens if we set $\gamma = 1$
(instead of 3)

ABRS "A_hat_1=0" "gamma=1"

Q2: What happens if we set $\ln \hat{A} = 0.2$

ABRS

Q3: What happens if we set $\ln \hat{A} = 0.2$ and
increase γ ? [▶ ABRs-toolkit \(web\)](#)



Questions II

Q4: What happens if we set $\ln \hat{A} = 0.2$ alone
vs. $\ln \hat{A} = 0.2$ and $\ln \hat{\eta} = 0.8$?

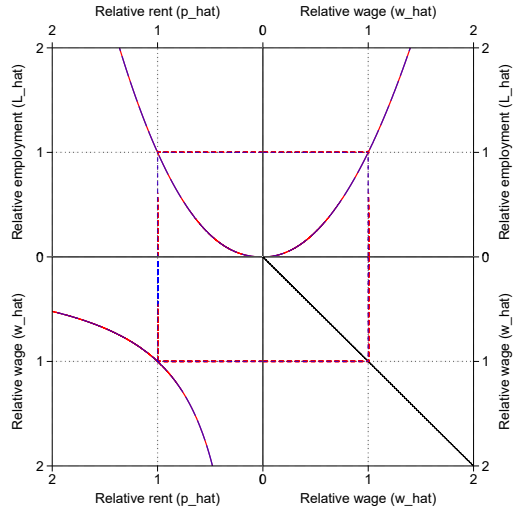
```
ABRS "A_hat_2=0.2" "eta_hat_2=0.8"
```

Q5: What happens if we set $\ln \hat{A} = 0.2$ alone
vs. $\ln \hat{A} = 0.2$ and $\ln \hat{\phi} = 0.8$?

```
ABRS "A_hat_2=0.2" "phibar_hat_2=0.8" "max_L_hat=4"
```

Q6: What happens if we keep $\ln \hat{A} = 0$ and set $\ln \hat{\eta} = 0.8$ and $\ln \hat{\varphi} = 0.8$?

```
ABRS "A_hat_1=0" "A_hat_2=0" "eta_hat_2=0.8" "phibar_hat_2=0.8"
"max_L_hat=4"
```



Conclusion

Summary

- ▶ With **extreme-value distributed location preferences**
 - ▶ Allows deriving discrete choice probabilities
 - ▶ Upward-sloping labor demand and downward-sloping housing demand
 - ▶ **More preference heterogeneity, less mobility** and steeper curves
- ▶ Important effects on general equilibrium See further comments on Q1-Q6 in Appendix
 - ▶ **Real wages no longer reflect QoL differences!** (unless γ is large)
 - ▶ Housing productivity increases real wages
 - ▶ Labour productivity increases real wages
 - ▶ **High real wages wrongly attributed to low QoL in RR!**

Next week: **The ARSW model**

Literature I

Core readings

- ▶ Ahlfeldt, G., Bald, F., Roth, D., Seidel, T. (2024): Measuring quality of life under spatial frictions: Toy version of the model. <https://github.com/Ahlfeldt/ABRS-toolkit>
- ▶ Ahlfedt, G., Redding, S., Sturm, D., Wolf, N. (2015): "Supplement to: The economics of Density: Evidence From the Berlin Wall" (Econometrica, 83(6)) , Sections S.2.1, S.2.2, S.2.3.
- ▶ Moretti, E. (2010): Local labour markets. Handbook of Labor Economics, Volume 4b.

Other readings

- ▶ Ahlfeldt, G. M., Bald, F., Roth, D., Seidel, T. (2026): [The urban equality of life premium](#). (previous version as CEPR DP19769).
- ▶ Armington, P. (1969). A Theory of Demand for Products Distinguished by Place of Production. IMF Staff Papers, 16(1), 159-178.
- ▶ McFadden, D. (1974): Conditional Logit Analysis of Qualitative Choice Behavior. In P. Zarembka (Ed.), Frontiers in Econometrics (pp. 105-142). Academic Press
- ▶ Saiz, A. (2010): The Geographic Determinants of Housing Supply. The Quarterly Journal of Economics, 125(3), 1253–1296

Appendix

Comments on Q1-Q6

- ▶ Q1
 - ▶ Curve in Quadrant 1 flatter since labor supply is less elastic
 - ▶ Notice that axes are reversed relative to the standard demand-supply diagram
 - ▶ Flatter curve implies **less** elastic labour supply
- ▶ Q2
 - ▶ Curve in Quadrant 1 is steeper since greater labor supply for the same wage
 - ▶ Consequentially, wage lower
 - ▶ Curve in Quadrant 2 steeper since for the same employment, wage is lower $\Rightarrow$ housing demand lower
 - ▶ Rent higher since the positive employment effect dominates the negative wage effect on housing demand
 - ▶ Real wage lower by 0.09 log units, less than half of the QoL difference (0.2)
 - ▶ Rosen-Roback framework delivers wrong measurement of QoL

Comments on Q1-Q6

- ▶ Q3
 - ▶ The larger γ , the closer the inverse real wage difference gets to the QoL difference
 - ▶ At $\gamma = 10$, real wage is 0.15 lower; at $\gamma = 10$, real wage is 0.175 lower; at $\gamma = 65$, real wage is 0.19 lower; at $\gamma = 500$, real wage is 0.199 lower
- ▶ Q4
 - ▶ Curve in the second Quadrant is steeper since lower rent due to greater housing supply
 - ▶ Curve in the first Quadrant is steeper since lower rent implies that workers accept lower wage
 - ▶ For a given employment level
 - ▶ Curve in the third Quadrant shifts inwards since at the same rent there are more workers $\Rightarrow$ lower wage clears the market

Comments on Q1-Q6

- ▶ Q5
 - ▶ Curve in Quadrant 5 shifts outwards since
 - ▶ labor demand increases as workers are more productive
 - ▶ at same rent (employment), firms pay higher wages (perfect competition)
 - ▶ RR **qualitatively wrong** since real wage difference positive
 - ▶ (implying negative QoL differential)
- ▶ Q6
 - ▶ Effects in first and second Quadrants like in Q4 ($\hat{\eta}$ effect)
 - ▶ Shift effects of $\hat{\eta}$ and $\hat{\phi}$ cancel out each other in Quadrant 3
 - ▶ Curve does not shift
 - ▶ RR **qualitatively wrong** since real wage difference positive
 - ▶ (implying negative QoL differential)